

Homework 2

Math 2031 — Professor Hamilton

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Section 2.1

Question 21

Statement: Let A and B be sets contained in universal set U . Which one(s) of the following statements are equivalent to the statement $A \subseteq B$? Give reasons for your answers.

- (A) $\forall x \in U, x \in A \wedge x \in B$. True by definition of being a subset.
- (B) $\forall x \in U, x \notin B \rightarrow x \notin A$. True, since A is a subset of B , if an element does not exist within B then we must conclude that it does not also exist within A .
- (C) $\exists x \in U \mid x \in A \wedge x \in B$. True by definition, since A is a subset of B , $x \in A$ and $x \in B$ are both true by definition, making the broader statement true as well.
- (D) $\forall x \in U, x \in A \rightarrow x \in B$. True by definition, since A is a subset of B , if $x \in A$ it must follow $x \in B$ by definition.
- (E) $\forall x \in U, x \notin A \rightarrow x \notin B$. False, let $A = \{1, 2, 3\}$ and $B = \mathbb{Z}$. $4 \in B \wedge 4 \notin A$ even though $A \subseteq B$.

Section 2.2

Question 1

Given: Let $A = \{1, 2, 3, 4, 5\}$, $B = \{0, 1, 4, 8\}$ and $C = \{2, 5, 7, 9, 11, 13, 17\}$. Compute each of the following:

- (A) $A \cup B = \{0, 1, 2, 3, 4, 5, 8\}$
- (B) $A \cap B = \{1, 4\}$
- (C) $A \cap C = \{2, 5\}$
- (D) $A - B = \{2, 3, 5\}$
- (E) $A - (B \cup C) = \{3\}$
- (F) $(A - B) \cup (A - C) = \{1, 2, 3, 4, 5\}$
- (G) $A \cap (B \cup C) = \{1, 2, 4, 5\}$
- (H) $(A \cap B) \cup (A \cap C) = \{1, 2, 4, 5\}$

- (I) $A \cup (B \cap C) = \{1, 2, 3, 4, 5\}$
 (J) $(A \cup B) \cap (A \cup C) = \{1, 2, 3, 4, 5\}$

Question 2

Given: Let A, B and C be the sets given in the previous exercise and let the universal set $U = \{0, 1, 2, 3, 4, \dots, 20\}$. Compute the following:

Note: Here ‘...’ means sequential integers from a to b .

- (A) $\overline{A} = \{0, 6, 7, 8, \dots, 20\}$
 (B) $\overline{B} = \{0, 2, 3, 5, 6, 7, 9, \dots, 20\}$
 (C) $\overline{A \cap B} = \{0, 2, 3, 5, \dots, 20\}$
 (D) $\overline{A \cup B} = \{0, 2, 3, 5, 6, 7, \dots, 20\}$
 (E) $\overline{A \cup B} = \{6, 7, 9, \dots, 20\}$
 (F) $\overline{A \cap B} = \{0, 6, 7, 10, \dots, 20\}$

Question 14B

Statement: Let A and B be sets. Prove the following: $A \cup (A \cap B) = A$

Proof:

Let A and B be sets and assume $x \in A \cup (A \cap B)$. We must show that:

I.) $A \cup (A \cap B) \subseteq A$ and II.) $(A \subseteq A \cup (A \cap B))$

Part I:

Let $x \in A \cup (A \cap B)$. We must show that $x \in A$ for the definition of a subset. Since $x \in A \cup (A \cap B)$ by assumption, $x \in A \vee (x \in A \wedge x \in B)$, which necessitates that $x \in A$ in both cases, thus we have shown $\forall x \in A \cup (A \cap B), x \in A$.

Part II:

Let $x \in A$. We must also show that $x \in A \cup (A \cap B)$ for the definition of a subset. By assumption $x \in A$ and as per Part I, $x \in A \cup (A \cap B)$ necessitates that $x \in A \therefore \forall x \in A, x \in A \cup (A \cap B)$.

Thus $A \cup (A \cap B) = A$. ■

Question 17

Statement: Let A, B , and C be sets.

- (A) **Conjecture:** $A \subseteq B \cup C \rightarrow A \subseteq B \vee A \subseteq C$

Proof:

Let A, B, C be sets such that $A \subseteq B \cup C$.

We want to show that $A \subseteq B \vee A \subseteq C$, i.e., $\forall x \in A, x \in B \vee x \in C$, by the definition of subset and union.

By assumption since $A \subseteq B \cup C$, it follows that $\forall x \in A, x \in B \vee x \in C$, as per the definition of union.

Case 1: $x \in B$

If we assume $x \in B$ by the definition of union, and since $x \in A$ by our initial assumption, it must follow that $A \subseteq B$ which would satisfy one of the conditions of our conclusion, which is what we wanted to demonstrate.

Case 2: $x \in C$

Similar behavior to the case above.

Therefore, we have shown that $A \subseteq B \cup C \rightarrow A \subseteq B \vee A \subseteq C$. ■

(B) $A \not\subseteq B \wedge A \not\subseteq C \rightarrow A \not\subseteq B \cup C$